link *bridging*

introduction to *network analysis* (*ina*)

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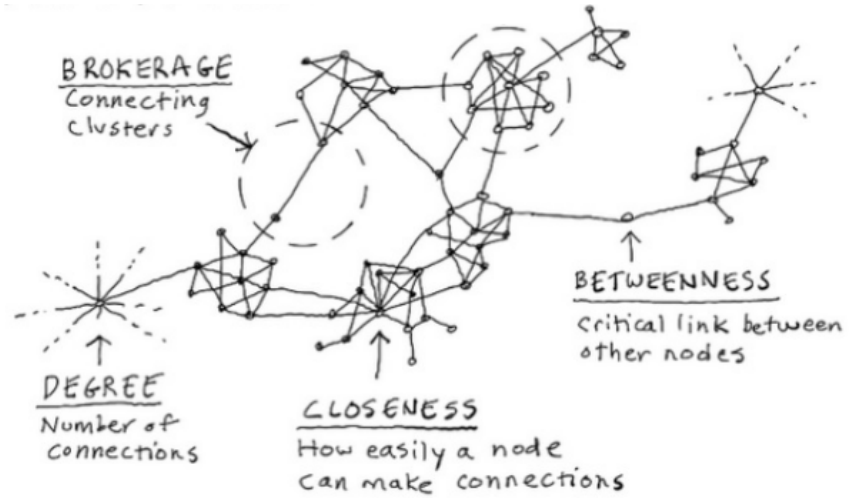
bridging *measures*

which *links* are most *important*?

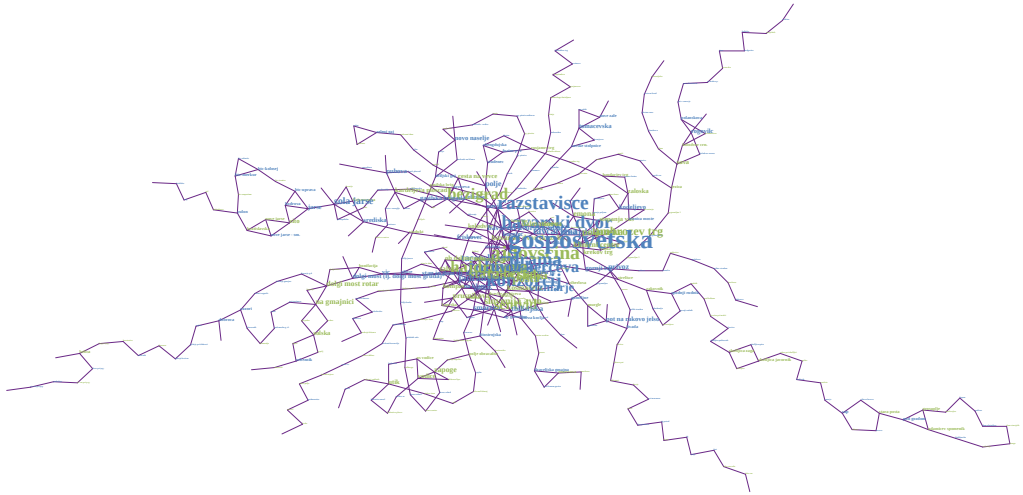
— *link bridging measures* for (*un*)*directed* networks
  – *betweenness-based* centrality [Fre77, FBW91, New05]

— *link embeddedness measures* for (*un*)*directed* networks
  – *topological overlap* measures [RSM+02, OSH+07, dNMB11]



networkology *LPP*

- partial *LPP public bus transport network**
- $n = 416$ bus stops with $\langle k \rangle = 2.72$ connections
- *giant component* 95.4% nodes (6 components)
- *"small-world"* with $\langle C \rangle = 0.09$ and $\langle d \rangle = 14.26$
- *"scale-free"* with $\gamma = 2.43$ for cutoff $k_{min} = 2$



* reduced to largest connected component of simple undirected graph

bridging *betweenness*

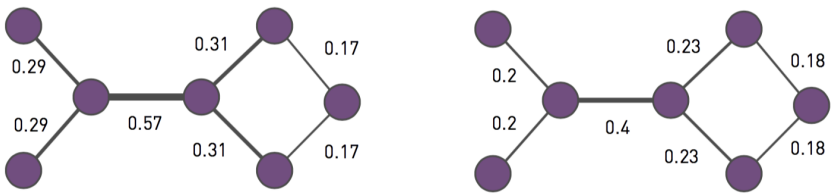
important *links* are *between other nodes*

— for (*un*)*directed* G *link betweenness* σ [Fre77] of $\{i, j\}$ is

- g_{st} is number of *geodesic paths between* s and t
- g_{st}^{ij} is number of *such geodesic paths through* $\{i, j\}$

$$\sigma_{ij} = \sum_{st \notin \{i,j\}} \frac{g_{st}^{ij}}{g_{st}}$$

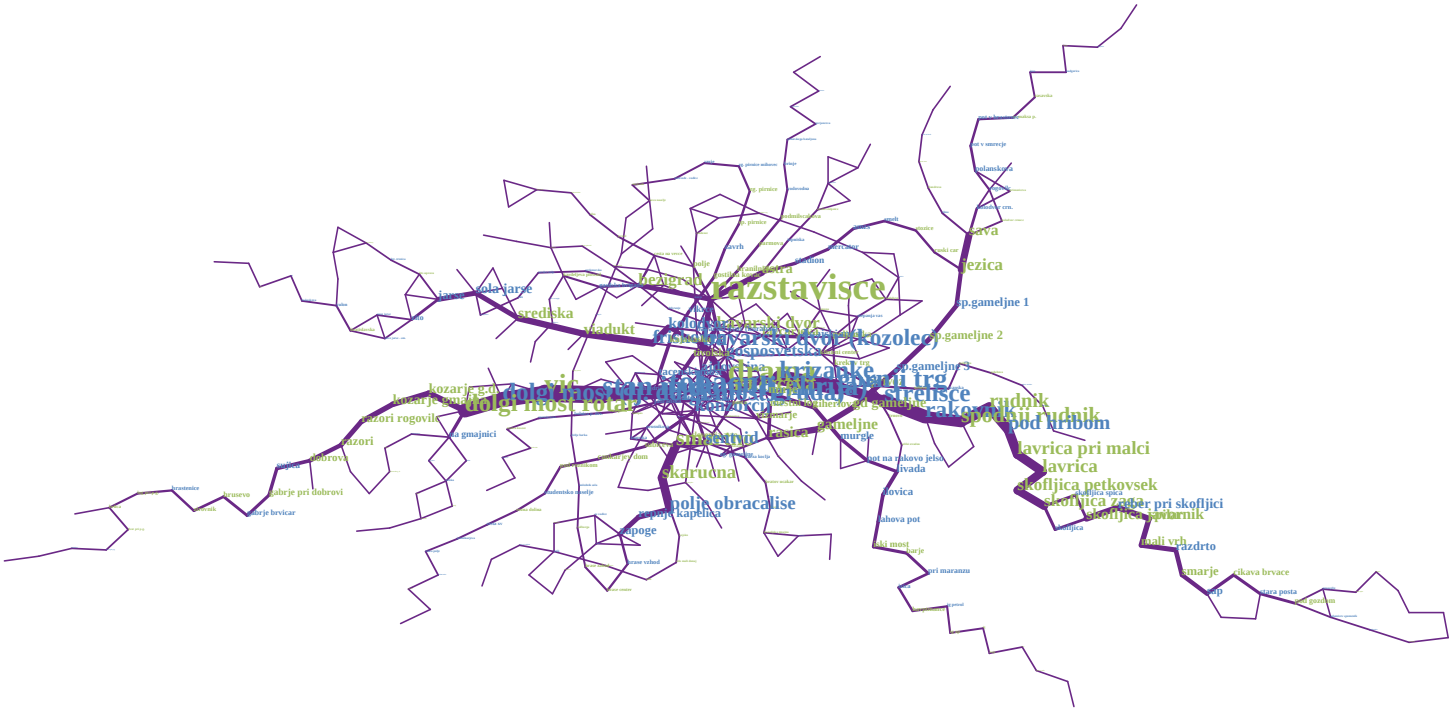
— σ considers *only geodesic paths* [FBW91, New05]



networkology *betweenness*

— *link betweenness* σ in partial LPP network†

— *highest* $\sigma_{ij} = 0.176n^2$ link is $\{$*Vič, Stan in dom*$\}$


razstavisce
kozolec
gameljne
lavrica pri malci
lavrica
skofljica petkovsek
skofljica zaver pri skofljici
skaruna
polje obracalise
jezica
gameljne 1
gameljne 2
gameljne 3
rudnik
pod urbom
dobrova
kozarje
kozori rogovilc
fabrje pri dobrovi
fabrje brvicar
krasnice
krasnevo
bezigrad
kolodvorska
vodovodni trg
streliscce
rakovnik
kapelica
apoge
kozolec
gmajnici
kozarje g.d.
kozarje
kozori
kozolec
kozolec
kozolec

bridging *bridgeness*

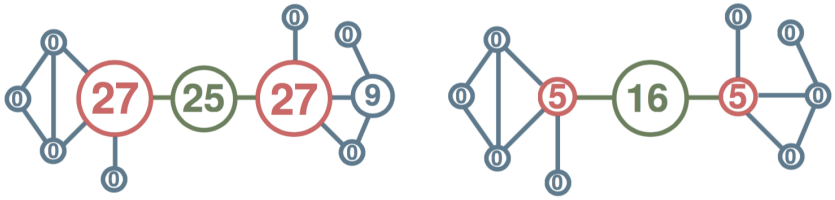
important *links* are *bridges between nodes*

— for (*un*)*directed* G *link bridgeness* $\widetilde{\sigma}$ [JMK+16] of $\{i, j\}$ is

- g_{st} is number of *geodesic paths between* s and t
- g_{st}^{ij} is number of *such geodesic paths through* $\{i, j\}$

$$\widetilde{\sigma}_{ij} = \sigma_{ij} - \sum_{st \in \Gamma_i \cup \Gamma_j} \frac{g_{st}^{ij}}{g_{st}} = \sum_{st \notin \Gamma_i \cup \Gamma_j} \frac{g_{st}^{ij}}{g_{st}}$$

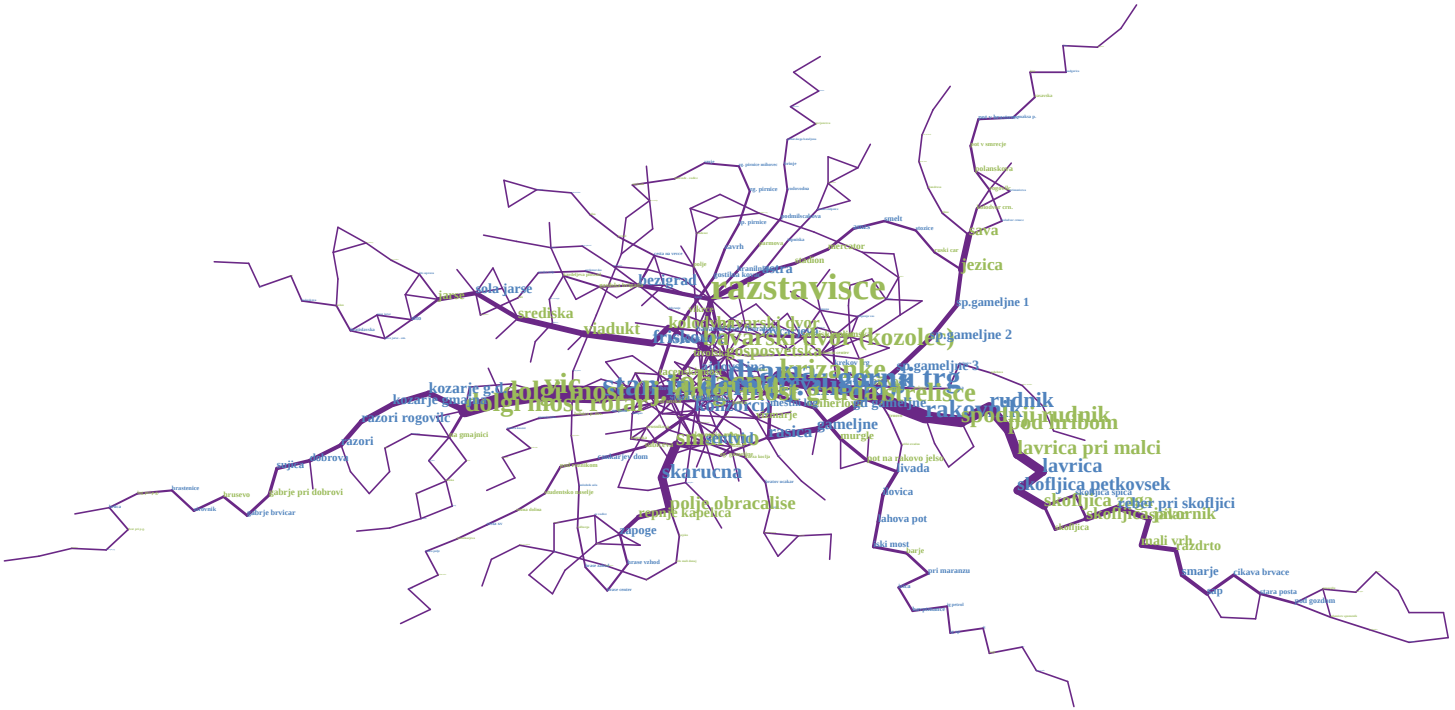
— σ mixes *local centers* with *global bridges* [JMK+16]



networkology *bridgeness*

— *link bridgeness* $\widetilde{\sigma}$ in partial LPP network[‡]

— *highest* $\widetilde{\sigma}_{ij} = 0.169n^2$ link is $\{$*Vič, Stan in dom*$\}$


razstavisce
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kozori rogovilc
kozori
krasnovo
krasnje pri dobrovi
krasnje brvicar
krasnice
kresnik
kresnjevo
kresnje
kresnje kapelica
kapoge
kapelica
kapelica veland
kapelica
kapelica
kapelica

bridging *embeddedness*

important *links* are *embedded between nodes*

— for *undirected* G *link embeddedness*[§] θ [OSH+07] of $\{i,j\}$ is

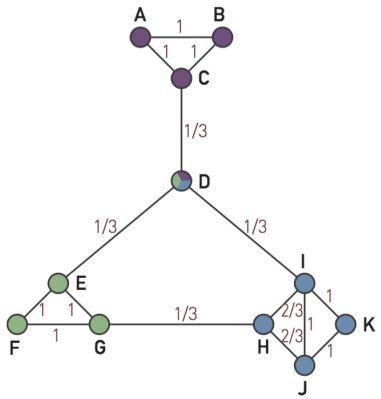
– Γ_i is set of *neighbors* or *neighborhood* of i

$$\theta_{ij} = \frac{|\Gamma_i \cap \Gamma_j|}{k_i - 1 + k_j - 1 - |\Gamma_i \cap \Gamma_j|} \qquad \theta_{ij} = 0 \text{ for } k_i = k_j = 1$$

— μ*-corrected link embeddedness* $\widetilde{\theta}$ [Bat19] of $\{i,j\}$ is

– μ is *maximum* number of *triangles* over *links*

$$\widetilde{\theta}_{ij} = \frac{|\Gamma_i \cap \Gamma_j|}{\mu + \max(k_i, k_j) - 1 - |\Gamma_i \cap \Gamma_j|}$$

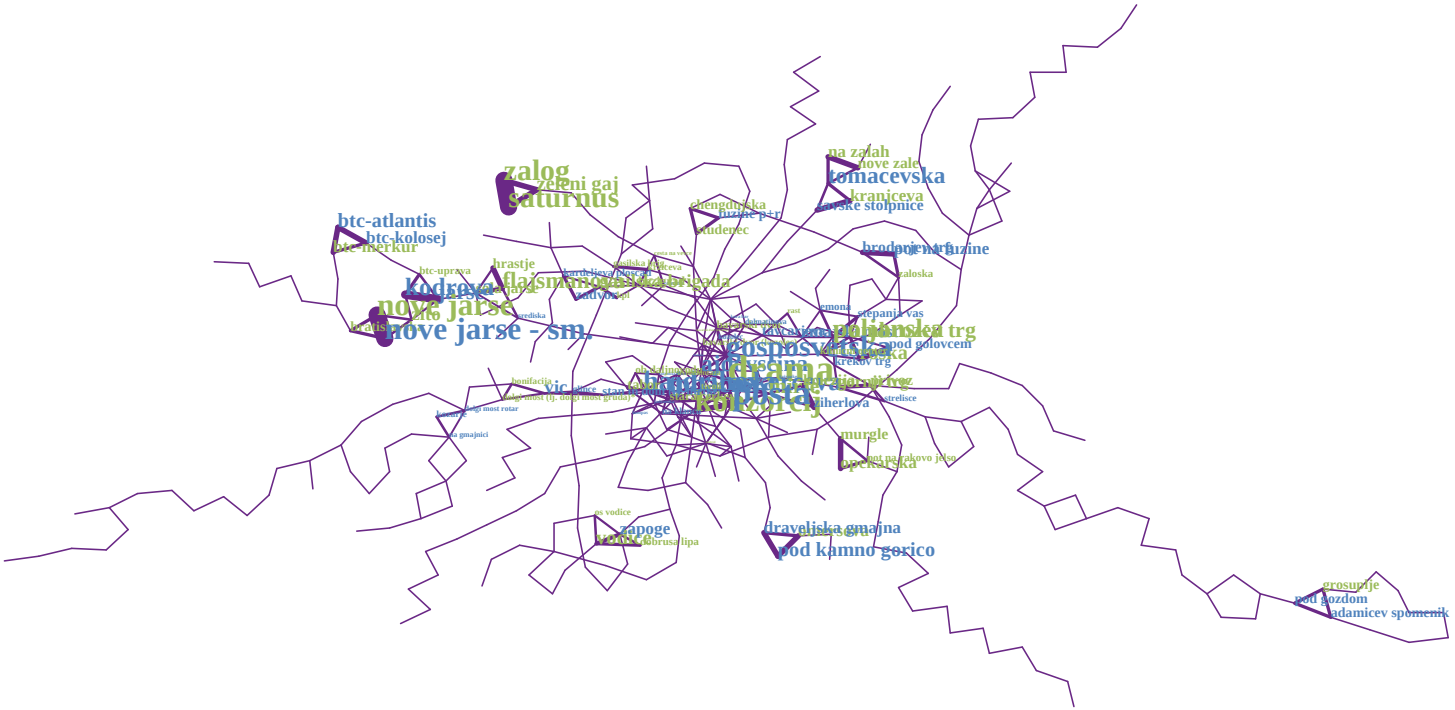


[§] θ & $\widetilde{\theta}$ better known as topological overlap indices/weights

networkology *embeddedness*

— *link embeddedness* θ in partial LPP network[¶]

— *highest* $\theta_{ij} = 1.0$ links are $\{$*Zalog, Saturnus*$\}$ *etc.*

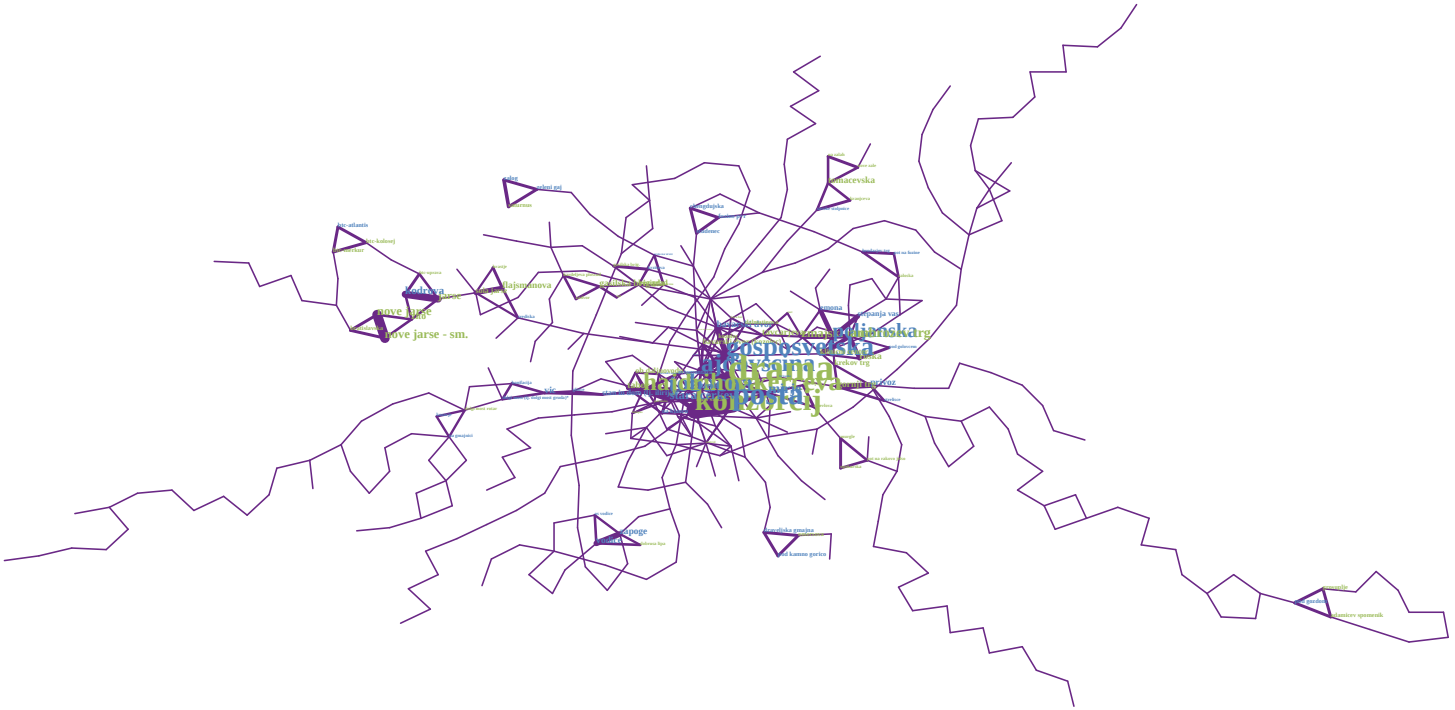


[¶] reduced to largest connected component of simple undirected graph

networkology *μ-embeddedness*

— *μ-corrected embeddedness* $\widetilde{\theta}$ in partial LPP network[‖]

— *highest* $\widetilde{\theta}_{ij} = 0.4$ links are $\{$*Pošta, Konzorcij*$\}$ *etc.*



‖ reduced to largest connected component of simple undirected graph

bridging *overview*

which *links* are most *important*?

	1 IA	2 IA	3 IIA	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 14 **DC** Degree Centrality																	2 8 **EC** Eigenvector Centrality
2	3 12 **BC** Betweenness Centrality	4 13 **CC** Closeness Centrality											5 9 **SC** Subgraph Centrality	6 4 **C_{COEF}** Clustering Coefficient	7 1 **$C_{COEF^{-1}}$** inverse CCOEF	8 1 **MNC** max. neighb. comp.	9 3 **EC_{COEF}** edge clustering coefficient	10 2 **PR** PageRank
3	11 2 **RL** RangeLimited Betweenness	12 8 **IC** Information Centrality											91 1 CCxxx	14 2 **SC_O** odd Subgraph Centrality	15 3 **LAC** loc. avg. Connectivity	16 3 **DMNC** dens. max. neighb. comp.	17 2 **SEC_{COEF}** sum of ECCOEF	18 1 **LR** LeaderRank
4	19 4 **BN** BottleNeck Centrality	20 2 **RC** Radiality Centrality	21 1 **IG** Integration	57 1 DCxx	58 1 BCxx	59 1 CCxx	61 1 ECCxx	62 1 KSxx	63 1 PRxx	64 1 IGxx	65 1 RCxx	89 1 DCxxx	90 1 BCxxx	32 2 **SC_E** even Subgraph Centrality	33 1 **KL** Clique Level	34 1 **COC_{COEF}** coexpr. weight CCOEF	35 2 **PEC_{coeff}** PCC×ECCOEF	36 2 **KS** KatzStatus
5	37 1 **RWBC** RandomWalk Betweenness	38 1 **RWCC** RandomWalk Closeness	39 1 **$CC_{2,3,4}$** 2,3,4-localized-CC	93 1 ECCxxx	95 1 PRxxx	94 1 KSxxx	50 1 CCC_{COEF}	97 1 RCxxx	96 1 IGxxx	46 1 DCBC	47 1 BCCC	48 1 CCKS	49 1 KSPR	50 1 DCPR	51 1 **β** Bipartivity	52 1 **SC_2** 2-localized-SC	53 5 **NC** Neighborhood Centrality	54 1 **EC_2** 2-localized-EC
6	55 2 **σ** stress Centrality	56 2 **ECC** Eccentricity	57-71 1 **WDC** Weighted Degree	72 1 DCECC	73 1 CCECC	74 1 BCECC	75 1 KSECC	76 1 PRECC	77 1 IGECC	78 1 DCCC	79 1 BCKS	80 1 CCPR	81 1 KSIG	82 1 DCIG	50 1 DCC_{COEF}	84 1 **SC_3** 3-localized-SC	85 1 **LI** Lobby Index	86 1 **EC_3** 3-localized-EC
7	87 1 **$BC_{2,3,4}$** 2,3,4-localized-BC	88 1 **ECC^{-1}** inverse Eccentricity	89-103 1 **SDC** Sphere Degree Centrality	104 1 DCRC	105 1 CCRC	106 1 BCRC	107 1 KSRC	108 1 PRRC	109 1 IGRC	110 1 DCKS	111 1 BCPR	112 1 CCIG	113 1 DCPR	114 1 BCIG	115 1 ECCRC	50 1 BCC_{COEF}	117 1 **SC_4** 4-localized-SC	118 1 **EC_4** 4-localized-EC

22 2 **FC** Functional Centrality	23 2 **FD** Functional Diversity	24 2 **US** UniScore	25 1 **DIS** Pairwise Dis-connectivity	26 1 **ASS** Assortative Mixing	27 1 **DAM** Damage	117 1 **UC** United compl. Centrality
28 1 **EI** Essentiality Index	29 1 **CM** Complexity Measure	30 1 **NαC** Normalized α Centrality	31 1 **MC** Moduland Centrality	13 1 **HGI** Harary Graph Information	116 0 **HYP** Hyperbolic Index	118 1 **HC** Harmonic Centrality

Z mass **C** Name

Hybrid

- Betweenness-based
- Distance-based
- Linear Combinations
- Subgraph-based
- Clustering Coefficient-based
- Edge Clustering Coefficient-based
- Spectral-based
- Miscellaneous

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bridging *references*

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bridging *references*

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